BOUNDING THE RELATIVE DIMENSION FROM A DENSE OPEN, FOR A FLAT MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The open subscheme of the source on which the morphism has bounded fibre dimension, and its decomposition into pieces. More on Morphisms, labels more-morphisms-lemma-flat-finite-presentation-CM-open and more-morphisms-lemma-flat-finite-presentation-CM-pieces, invoked at proof lines 3035-3036. Provides the open subscheme containing every generic point of every fibre and its decomposition by the fibre dimension, which reduces the statement to the vanishing of the pieces of dimension larger than the bound.
- Openness of a flat locally finitely presented morphism, and the dense open case with retrocompactness. Morphisms, label morphisms-lemma-fppf-open (proof line 3042); Same chapter flat.tex, label lemma-flat-finite-type-finitely-presented-over-dense-open-X (proof line 3049). The first makes the image of each piece open so that it must meet the given dense open, and the second turns the retrocompact dense open case into the locally finitely presented case.

Source reference. Original lemma 081P: sources/stacks-flat/pinned-original-source.tex, source lines 3000–3000. Navigation pointer for \ref{lemma-flat-finite-type-finitely-presented-over-dense-open-X} used inside the counted span (line 3049); the argument that consumes it is retained in the span.

Source reference. Original lemma 045U: sources/stacks-chow/tags.txt, source lines 5449–5449. Navigation pointer for \ref{more-morphisms-lemma-flat-finite-presentation-CM-open} used inside the counted span (line 3036); the argument that consumes it is retained in the span.

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Source reference. Original lemma 054T: sources/stacks-chow/tags.txt, source lines 6651–6651. Navigation pointer for \ref{more-morphisms-lemma-flat-finite-presentation-CM-pieces} used inside the counted span (line 3037); the argument that consumes it is retained in the span.

Source reference. Original definition 02NJ: sources/stacks-chow/tags.txt, source lines 3482–3482. Navigation pointer for \ref{morphisms-definition-relative-dimension-d} used inside the counted span (line 3024); the argument that consumes it is retained in the span.

Source reference. Original lemma 01UA: sources/stacks-chow/tags.txt, source lines 2403–2403. Navigation pointer for \ref{morphisms-lemma-fppf-open} used inside the counted span (line 3042); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes which is flat and locally of finite type. Let $U \subset S$ be a dense open such that $X_U \to U$ has relative dimension $\leq e$, see Morphisms, Definition 02NJ. If also either*

(1) *f is locally of finite presentation, or*
(2) *$U \subset S$ is retrocompact,*

then f has relative dimension $\leq e$.

Proof. Proof in case (1). Let $W \subset X$ be the open subscheme constructed and studied in More on Morphisms, Lemmas 045U and 054T. Note that every generic point of every fibre is contained in W, hence it suffices to prove the result for W. Since $W = \bigcup_{d \geq 0} U_d$, it suffices to prove that $U_d = \emptyset$ for $d > e$. Since f is flat and locally of finite presentation it is open hence $f(U_d)$ is open (Morphisms, Lemma 01UA). Thus if U_d is not empty, then $f(U_d) \cap U \neq \emptyset$ as desired.

Proof in case (2). We may replace S by its reduction. Then U is scheme theoretically dense. Hence f is locally of finite presentation by Lemma 081P. In this way we reduce to case (1). □